
Algorithm 2 Frame Difference Detector (FrameDiffDet)

Require: Downsampled frames $\tilde{f}_t, \tilde{f}_{\text{prev}}$

Require: τ_d : pixel difference sensitivity threshold

Ensure: $\mathbf{M}_t \in \{0, 255\}$: binary foreground mask

Convert $\tilde{f}_t$ and $\tilde{f}_{\text{prev}}$ to grayscale: F_t, F_{prev}

2: Compute binary mask, where each pixel (i, j) is set as:

$$\mathbf{M}_t(i, j) \leftarrow \begin{cases} 255 & \text{if } |F_t(i, j) - F_{\text{prev}}(i, j)| > \tau_d \\ 0 & \text{otherwise} \end{cases}$$

return $\mathbf{M}_t$
